

Influence of Ni addition on strain-induced martensite in austempered ductile iron

Xin Wang^a, Yuzhou Du^{a,*}, Ziyu Wang^b, Haohao Li^a, Yanyin Zheng^b, Pengchun Li^a, Tianjian Mi^a, Bailing Jiang^a

^a School of Materials Science and Engineering, Xi'an University of Technology, Xi'an, 710048, PR China

^b Xi'an Technology of Eutectic Metal Co., Ltd., 710086, PR China

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ABSTRACT

This study systematically investigates the effects of nickel (Ni) additions on strain-induced martensitic transformation in austempered ductile iron (ADI). The matrix of ADI containing no more than 4 wt% Ni was composed of acicular ferrite and austenite. Increasing Ni content elevated the austenite volume fraction but reduced its carbon concentration. Furthermore, Ni additions suppressed the nucleation rate of the α -phase, resulting in an increase in both the quantity and size of blocky austenite as Ni content was increased. Strain-induced martensite was observed within the blocky austenite for ADI containing 2 wt% and 4 wt% Ni. Notably, approximately 40 % of the austenite in the ADI containing 4 wt% Ni transformed into martensite following the tensile test. This substantial phase transformation effectively mitigated stress concentrations and significantly enhanced the work-hardening capacity. Moreover, concurrent deformation twinning was observed within the blocky austenite. Consequently, an exceptional strength-ductility balance with an ultimate tensile strength of ~ 1100 MPa and a total elongation of ~ 10 % was achieved.

1. Introduction

Austempered ductile iron (ADI) is extensively used in various industries because of its superior mechanical properties, good machinability, and low cost [1,2]. ADI's matrix is characterized by acicular ferrite (α) and high-carbon austenite (γ) [3]. Currently, the mechanical properties of ADI are mainly regulated by optimizing austempering parameters to control the size and relative fraction of α/γ phase [4]. For instance, a refined ausferrite microstructure was obtained by austempering at 230 °C, yielding a high ultimate tensile strength (UTS) of 1420 MPa but a limited total elongation (TL) of 1.3 %; conversely, raising the austempering temperature to 380 °C significantly increases the austenite fraction, which gives rise to a high TL of 3.9 % but reduces UTS to 979 MPa [5]. In general, the elongation of ADI is usually less than half that of forged steels with similar strength levels [6]. Therefore, improving the ductility of ADI is critical for expanding its applications. We previously made efforts to enhance the ductility of ADI by adjusting the carbon distribution between the α and γ phases [7]. Nonetheless, when the UTS of ADI exceeds 1200 MPa, its elongation is still less than 6 %. Further increasing the ductility of ADI remains challenging.

It should be noted that austenite in steels can consume strain energy and induce martensitic transformation during deformation, thereby improving their strength and ductility [8]. For instance, in 0.2C–1.6Al–6.1Mn–Fe steel, approximately 80 % of austenite undergoes martensitic transformation, achieving an outstanding product of strength and elongation (PSE) exceeding 40 GPa% [9]. Moreover, the austenitic matrix in Fe18Mn1.0C steel transforms into martensite during straining, resulting in a high UTS of 1147 MPa and a TL of 60.7 % [10]. Notably, strain-induced martensitic transformation has been observed in quench-partitioning (QP) and dual-phase steels [11], which contain approximately 20 % austenite. It is well known that a 20 %–50 % austenite fraction existed in ADI [12]. Although strain-induced martensite has also been observed [13], its limited volume fraction restricts mechanical properties. This is primarily attributed to the exceptionally high stability of ADI austenite, where its carbon enrichment exceeds 1.8 wt% [14]. Therefore, deliberate destabilization of austenite is essential to activate substantial TRIP effects and achieve further improvements in the mechanical properties of ADI.

The stability of austenite is closely related to its stacking fault energy (SFE). When the SFE is less than 15 mJ/mm², austenite preferentially

* Corresponding author.

E-mail address: duyuzhou@xaut.edu.cn (Y. Du).

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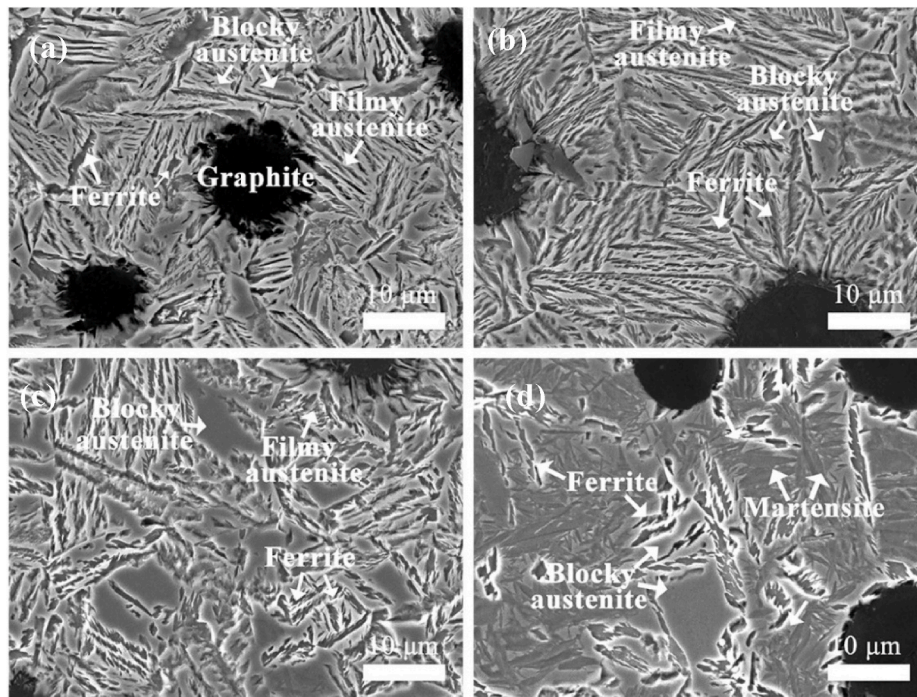


Fig. 1. SEM images of the (a) 0Ni, (b) 2Ni, (c) 4Ni, and (d) 6Ni samples.

transforms into martensite during deformation. In contrast, with an SFE ranging from 15 to 45 mJ/mm², austenite tends to form deformation twins. When the SFE exceeds 45 mJ/mm², dislocation slip predominantly occurs [15]. The SFE of austenite in ADI is commonly higher than 30 mJ/mm² due to its high carbon content; thus, strain-induced martensitic transformation is hard to occur [16]. Ni addition in carbon steel is reported to be capable of reducing SFE and promoting martensitic transformation by substituting Fe atoms [17]. It is reported that 1.5 wt% Ni addition in ADI escalates the probability of martensitic transformation [13]. We previously observed the strain-induced martensite phenomenon in a ductile iron containing 6 wt% Ni, which gave rise to an excellent ductility with an elongation of ~21 % [18].

Therefore, in order to utilize the strain-induced martensite to improve the mechanical properties of ADI, Ni was added and the effects of Ni addition on the strain-induced martensitic transformation were investigated, with the objective to provide new insights into effective strategies for ductility improvement of ADI.

2. Experimental procedure

In this study, ductile irons with varying Ni addition (0, 2, 4, 6 wt%) were fabricated using horizontal continuous casting. The detailed fabrication procedure has been previously reported [19]. The chemical compositions of the Ni alloyed samples were determined through a QSN-750 optical emission spectrometer, which was approximately: 3.4 wt% C, 2.5 wt% Si, 0.14 wt% Mn, 0.01 wt% S, 0.04 wt% P, 0.03 wt% RE, 0–10 wt% Ni and balance Fe. Based on their Ni content, the samples were denoted as 0Ni, 2Ni, 4Ni, and 6Ni samples in the following sections. The relationship among process parameters, microstructure, and mechanical properties of unalloyed ADI has been systematically investigated [12]. In order to obtain a high volume fraction of austenite, the austenitizing temperature was selected as 900 °C, and the austempering temperature was selected as 380 °C. The austempering duration, which governs carbon diffusion kinetics [20], was set at 90 min to ensure complete phase transformation while preventing carbide precipitation. Therefore, the as-cast ductile iron was initially heated to 900 °C and held for 120 min in a vacuum furnace, then the samples were rapidly transferred to a salt bath consisting of 50 % NaNO₂ and 50 % NaNO₃ at 380 °C

for 90 min, followed by air cooling to ambient temperature.

For microstructure analysis, the samples were prepared by mechanically grinding, polishing, and etching with 4 % nital solution. A VEGA 3 XMU scanning electron microscopy (SEM) with an accelerating voltage of 30 KV was used to characterize the microstructure. Further detailed microstructures were characterized via JEM-3010F transmission electron microscopy (TEM) with an accelerating voltage of 300 KV. The TEM samples were prepared by mechanical polishing, followed by twin-jet electropolishing in a solution of 95 % perchloric acid and 5 % ethanol, and then ion-milled to remove the oxide layer. X-ray diffraction (XRD) analysis was performed using an XRD-7000 diffractometer with Cu-K α radiation ($\lambda = 1.5406 \text{ \AA}$). The measurements were conducted over a 2θ range of 35°–95° with a step size of 4°/min. The austenite fraction was estimated via direct comparison from integrated intensities of (200) α , (211) α , (200) γ , (220) γ , and (311) γ diffraction peaks. Moreover, the carbon content of austenite was estimated using the following equation [21]:

$$a_{\gamma} = 0.3548 + 0.0044 C_{\gamma} \quad (1)$$

where a_{γ} represents the lattice parameters of austenite (nm), which were obtained via Nelson–Riley extrapolation; C_{γ} represents the carbon concentration of austenite (wt.%). In order to investigate the microstructure evolution, the MUCG83 thermodynamic model was applied to calculate the nucleation driving forces of the α phase during the austempering process [22].

The tensile tests were conducted on samples with a cross-section of 4 mm $\times$ 2 mm² and a gauge length of 24 mm, using an HT-2402 universal testing machine at a strain rate of $1.6 \times 10^{-3} \text{ s}^{-1}$. To ensure repeatability, at least three specimens were tested. Following mechanical testing, fracture surface morphology was characterized by SEM. Microhardness measurements of the matrix were performed with an HVT-1000 microhardness tester, applying a load of 100 g for 15 s. At least 10 measurements were taken, and their average was used as the final result.

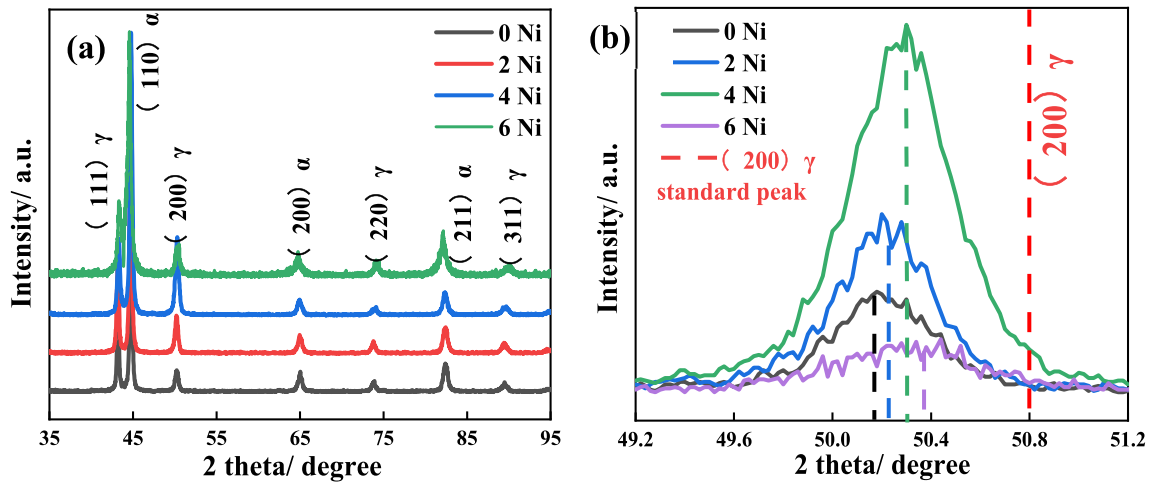


Fig. 2. (a) XRD patterns, (b) the enlarged (200) γ peaks of ADI samples with different Ni contents.

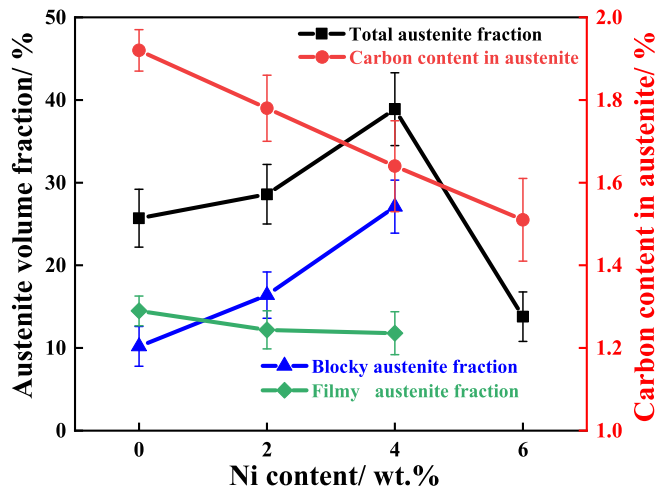


Fig. 3. The variations of austenite and its carbon content with Ni content.

3. Results

3.1. Effects of Ni addition on the microstructure of ADI

Fig. 1 presents the SEM images of ADI samples with different Ni additions. It could be seen that the microstructure varied with increasing Ni content. The matrix of 0Ni, 2Ni, and 4Ni samples mainly consisted of acicular ferrite (α phase) and austenite (γ phase). Austenite with blocky and filmy morphology could be detected, in which the blocky austenite was referred to as submicron-sized austenite trapped between ferrites. With increasing Ni addition, the austenite fraction increased, and the blocky austenite size increased from $\sim 2.4 \mu\text{m}$ for the 0Ni sample to $\sim 8.6 \mu\text{m}$ for the 4Ni sample. However, when the Ni addition increased to 6 wt%, the matrix was changed into martensite and retained austenite. Some coarse austenites with a size of $\sim 10.4 \mu\text{m}$ were also detected in the 6Ni sample (Fig. 1d).

Fig. 2 shows the XRD patterns of the ADI samples. The intensity of the γ phase initially increased and then decreased with increasing Ni content, indicating that its austenite fraction varied. When the Ni addition increased to 4 wt%, the (200) γ phase intensity reached its maximum, corresponding to a high austenite fraction. The standard (200) γ peak is also shown in Fig. 2b. It could be seen that the peak of the (200) γ phase gradually shifted to the right with increasing Ni content, indicating a reduction in carbon content within the austenite.

Based on the XRD analysis, the volume fraction of austenite and its

carbon content were evaluated, as shown in Fig. 3. The austenite fraction increased from 25.7 % for the 0Ni sample to 38.9 % for the 4Ni sample, while the austenite fraction was reduced to 13.8 % for the 6Ni sample, which was consistent with the SEM observations. Moreover, the volume fraction of blocky austenite was evaluated in SEM images, and then the filmy austenite fraction was calculated, which is also presented in Fig. 3. The volume fraction of blocky austenite increased from 10.2 % to 27.1 % with Ni increasing to 4 wt%, indicating that Ni addition facilitated the preservation of blocky austenite in ADI. Importantly, the carbon content in austenite continuously decreased with increasing Ni addition. When the Ni content increased to 4 wt%, the carbon content decreased from ~ 1.9 % to ~ 1.6 %, which would be beneficial to reduce the stability of austenite [23].

Fig. 4 shows the TEM images of ADI samples. A lamellar microstructure could be observed in 0Ni, 2Ni, and 4Ni samples, which was a typical microstructure of ADI [12]. The selected area electron diffraction (SAED) patterns confirmed that the bright and dark areas were the α and γ phases, respectively. Notably, with increasing Ni content, the α phase thickness and γ phase size increased. However, for the 6Ni sample, based on its morphological characteristics and SAED results, its matrix contained martensite and retained austenite (Fig. 4d).

The microstructure of ADI was determined by the nucleation and growth of the α phase [24]. The nucleation driving force of the α phase was closely related to the austempering temperature and alloy elements. In the present study, Ni content was the main factor influencing the nucleation of the α phase. The atomic radius of Ni ($R \sim 0.124 \text{ nm}$) and Fe ($R \sim 0.127 \text{ nm}$) were similar and thus Ni could substitute Fe atoms in the austenite lattice [25]. Moreover, Ni's 3d orbital interacted with Fe's 3d orbital and C's 2p orbital and enhanced the austenite stability [26]. Therefore, Ni addition would inhibit the transformation of the γ phase to the α phase. The nucleation driving force of the α phase for the samples during austempering treatment was calculated using the Mucg83 thermodynamic model, as shown in Fig. 5 [22]. It indicated that the nucleation driving force of the α phase rapidly decreased with increasing Ni content. Therefore, the microstructure was coarsened with increasing Ni addition [23].

Fig. 6 shows the schematics of the microstructure evolution for unalloyed and Ni-alloyed ADIs. During austempering treatment, α initially nucleated at the prior austenite grain boundaries or the austenite/graphite interface, and the newly formed α phase would provide additional nucleation sites in the following [27]. Ni addition significantly reduced the α phase nucleation driving force, resulting in a low α phase nucleation rate. In the unalloyed ADI, a high nucleation rate of the α phase would restrict its growth (Fig. 6a and b), resulting in a fine microstructure. Conversely, the Ni-alloyed sample exhibited a coarse

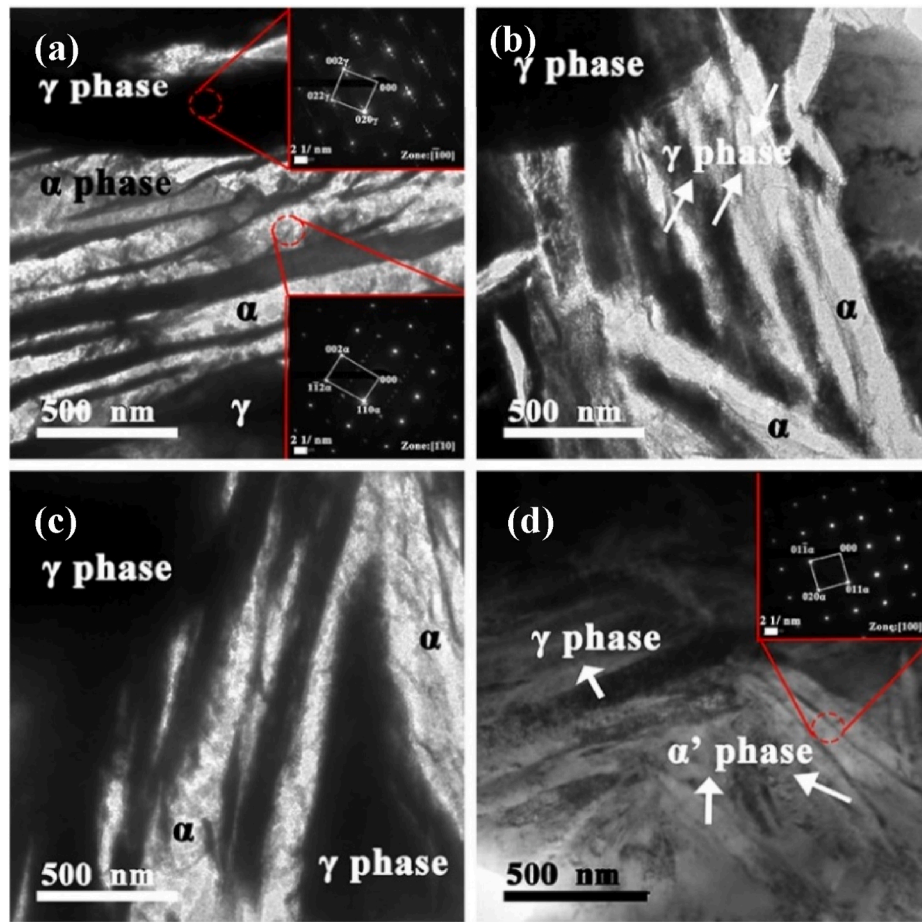


Fig. 4. TEM images of the (a) 0Ni, (b) 2Ni, (c) 4Ni, and (d) 6Ni samples.

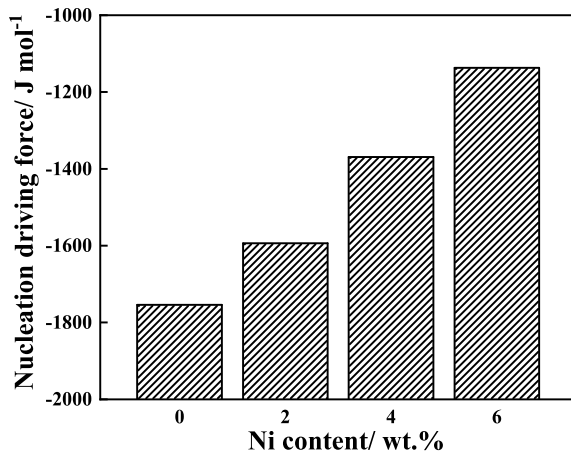


Fig. 5. The nucleation driving force of α phase for ADI samples with different Ni contents during austempering treatment at 380 °C calculated using the MUCG 83 code.

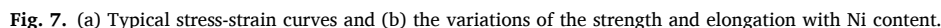
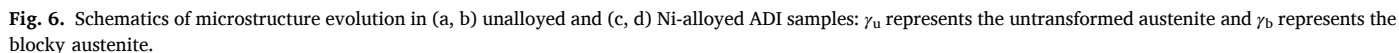
microstructure due to a low nucleation rate (Fig. 6c and d). After the formation of the α phase, carbon atoms diffused from the α phase to the surrounding γ phase, enhancing its stability and enabling it to be preserved at room temperature [28]. The α phase with a similar orientation and the thin filmy γ phase distributed between them formed ausferrites. The γ phase trapped ausferrites would be in a blocky shape. High α phase nucleation rate led to dense ausferrites and small blocky austenite (Fig. 6a and b). In contrast, Ni addition reduced the α phase nucleation

rate, allowing ausferrites to grow freely and retaining coarse blocky austenite (Fig. 6c and d). The Ni addition reduced the carbon content in austenite, which came from two sources: carbon atoms transferred from graphite during austenitizing and from the α phase during austempering. The Ni-Fe substitution increased the lattice distortion and reduced the solid solubility of carbon in prior austenite during austenitizing. Additionally, the few ferrite nuclei inhibited the diffusion of carbon atoms from the α phase to the γ phase [29]. Therefore, with increasing Ni content, the carbon content in austenite decreased. However, for the 6Ni sample, the lack of carbon supplementation in blocky austenite gave rise to low stability, causing it to transform into martensite during austempering.

3.2. Effects of Ni addition on martensitic transformation in ADI

3.2.1. Mechanical response

Fig. 7a shows the stress-strain curves of ADI samples with different Ni additions. The 0Ni, 2Ni, and 4Ni samples exhibited a good combination of strength and ductility, while no obvious ductile deformation was observed in the 6Ni sample due to the brittle martensitic matrix. The ultimate tensile strength (UTS) and total elongation (TL) obtained from the stress-strain curve are presented in Fig. 7b. The strength and ductility were enhanced when the Ni addition increased to 4 wt%. However, further increases in Ni content led to a sharp decline in ductility. Very interesting phenomenon that the strength and ductility were simultaneously improved with increasing Ni content, which was different from the commonly reported contradiction between strength and ductility [30]. When the Ni content increased to 4 wt%, the UTS increased from 933 MPa to 1102 MPa, the TL increased from 6.4 % to



The strength was closely related to work hardening behavior. The 4Ni sample exhibited a low YS/UTS ratio of ~ 0.7 , indicating a high work hardening capacity. Fig. 8 shows the variation of the work hardening rate with the strain of Ni-alloyed samples. The work hardening of all the ADIs decreased rapidly due to the ferrite deformation (Stage I) [31], then decreased slightly and accompanied by fluctuations (Stage II) attributed to the TRIP effect [32], and finally decreased again and ultimately led to fracture. The work hardening curve indicated that the 4Ni sample exhibited stronger work hardening capacity, and its

$$\sigma = K\varepsilon^n \quad (2)$$

where σ represents the true stress, K represents the strength coefficient,

where σ represents the true stress, K represents the strength coefficient, and ε represents the true Strain. A high value of n corresponds to a high work hardening rate. With increasing Ni content, the n increased from 0.10 for the 0Ni sample to 0.13 for the 2Ni sample, and then increased to 0.18 for the 4Ni sample. In contrast, the work hardening rate of the 6Ni sample decreased continuously until fracture, indicative of a poor work hardening capacity.

Fig. 9 shows the fracture surfaces of ADI samples. Graphite and graphite cavities were observed on the fracture surfaces. Dimples and cleavage facets could be seen in the 0Ni, 2Ni, and 4Ni samples (Fig. 9a–c), indicating a ductile–brittle mixed fracture. For the 6Ni sample, SEM analysis of fracture surfaces revealed exclusively cleavage

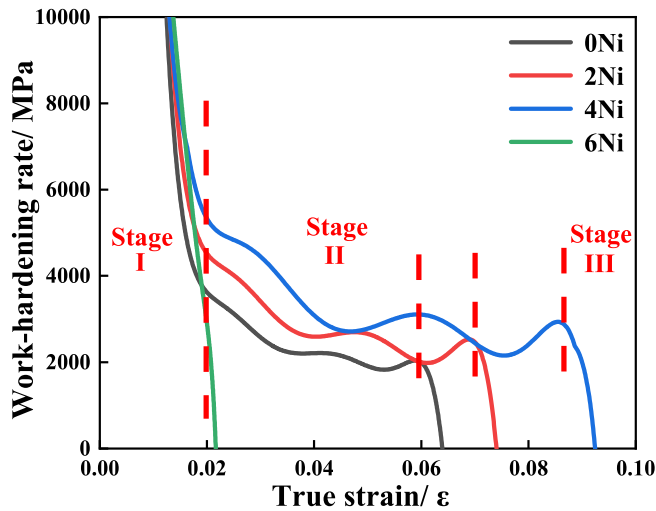


Fig. 8. The work hardening rate of ADI samples with different Ni contents.

facets (Fig. 9d), characteristic of brittle fracture in its martensitic matrix. This observation correlates well with the sample's limited ductility observed in tensile testing.

3.2.2. Strain-induced martensite phenomenon

In order to investigate whether austenite in the Ni-alloyed sample transformed into martensite during deformation, XRD analysis was performed on the fractured samples. The residual austenite content in the ADI samples was estimated, as shown in Fig. 10. The results revealed a decrease in austenite fraction in all Ni-alloyed samples, indicating that austenite had transformed into martensite. In addition, except for the 6Ni sample, the amount of austenite transformation increased with increasing Ni content. When the Ni content increased to 4 wt%, the transformed austenite increased from 3.8 % to 14.6 %, and the austenite transformation ratio increased from 15.3 % to 37.6 %. It indicated that nearly 40 % of the austenite had transformed into martensite in the 4Ni sample. For the 6Ni sample, there was almost no martensitic

transformation.

The SEM images of the samples after fracture are shown in Fig. 11. No obvious microstructural changes were observed in the 0Ni and 6Ni samples, but with increasing Ni content, newly formed martensite on the original blocky austenite was detected for the 2Ni (Fig. 11b) and 4Ni samples (Fig. 11c). This indicated that austenite transformed into martensite during deformation. Furthermore, more martensite was detected in the 4Ni sample (Fig. 11c), which was consistent with the XRD analysis. Microhardness measurements of the 4Ni sample's blocky austenite revealed an increase from 236.7 HV to 375.6 HV, further confirming the strain-induced martensitic transformation.

Fig. 12 presents the TEM images of the 4Ni samples after the tensile test. Only a small amount of austenite (A) remained in the 4Ni sample, but plenty of martensite (M) was detected (Fig. 12a). This confirmed that blocky austenite transformed into martensite during deformation in the 4Ni sample. The carbon content of large-scale martensite (Fig. 12a) was estimated to be about ~1.2 % through EDS point scanning (black region). The low carbon content in this region promoted the martensitic transformation of austenite. Large-scale martensite was surrounded by untransformed austenite. Some martensite was also observed in the surrounding austenite (Fig. 12b and c). High-resolution TEM (Fig. 12d) and corresponding selected area electron diffraction (SAED) results (Fig. 12e) indicated the untransformed austenite had a K-S orientation relationship of $(-111) \gamma // (0-11) \alpha$ and $[0-11] \gamma // [-1-1-1] \alpha$ with adjacent martensite. EDS mapping analysis implied that Si and Ni were distributed across martensite and austenite homogeneously, while austenite contained more carbon atoms (Fig. 12g-f). In order to determine the distribution of carbon concentration across austenite and martensite, an EDS line scan was performed, and the result is shown in Fig. 12f. The carbon concentration in austenite and martensite was about ~1.6 wt% and ~1.3 wt%, respectively. It was known that the martensitic transformation proceeded through a shear mechanism without carbon atom diffusion [34]. Therefore, it could be inferred that the distribution of carbon content in the blocky austenite was inhomogeneous, and martensitic transformation was preferentially induced in the region with low carbon content. It was worth noting that some twinning was observed in the undeformed austenite (Fig. 12a).

Additionally, some twinning was observed in the 4Ni sample after

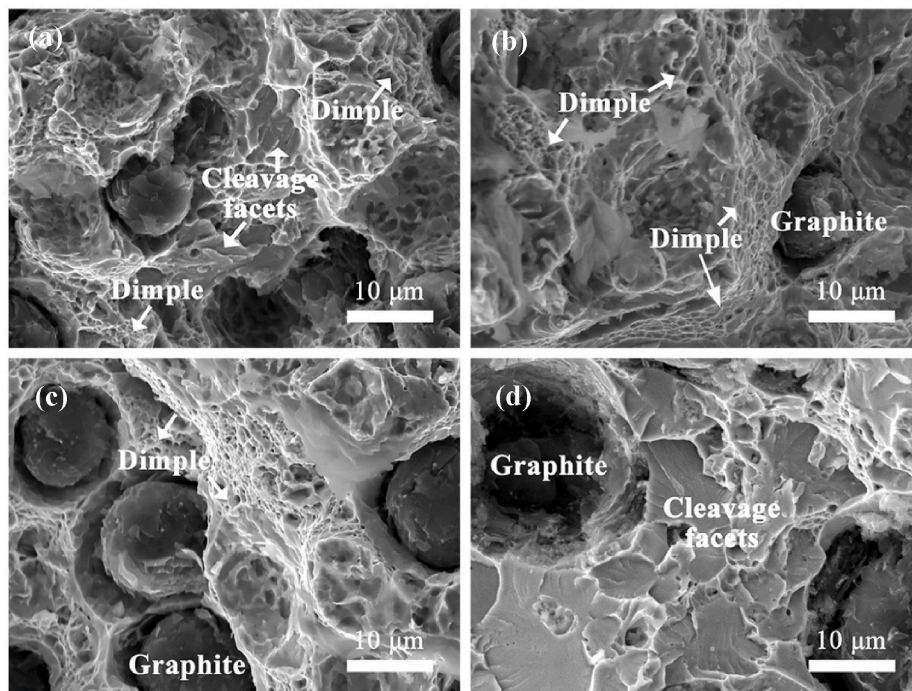


Fig. 9. SEM images of the fracture surfaces of (a) 0Ni, (b) 2Ni, (c) 4Ni, and (d) 6Ni samples.

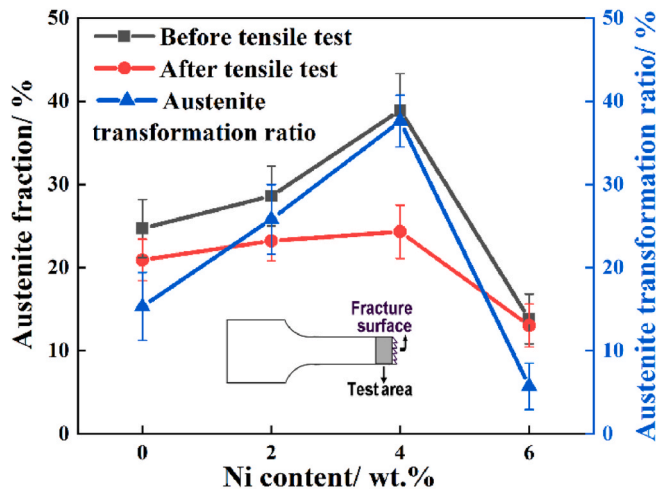


Fig. 10. The variations of austenite volume fraction with Ni content before and after fracture.

the tensile test, as shown in Fig. 13, indicating that twinning deformation occurred. It was known that the deformation mode of austenite was closely related to its stacking faults, predominantly governed by the content of interstitially dissolved carbon [15]. Consequently, the stacking fault in the blocky austenite varied due to the inhomogeneous distribution of carbon atoms (Fig. 12f). This variation led to twinning deformation in certain regions of the blocky austenite, attributable to its intermediate stacking fault.

4. Discussion

From the microstructure analysis, it could be seen that Ni addition promoted the increase of volume fraction and size of blocky austenite, which could transform into martensite and twins during deformation. Consequently, the work hardening rate was increased, and the strength and ductility were improved. The following will discuss the strain-

induced martensitic transformation in ADI.

In this study, a high austenite transformation ratio (~40 %) was observed in the 4Ni sample. However, for the 0Ni sample, strain-induced martensite was challenging. The stability of austenite directly determines the martensitic transformation, which is closely related to its chemical composition, size, morphology, and surrounding structure [35, 36]. It has been reported that the driving force of martensitic transformation can be estimated using the following equation [9]:

$$\Delta G^{\text{ch}} = -7381.6 + 69447X_{\text{C}} + 19296X_{\text{Mn}} - 38776X_{\text{C}}X_{\text{Mn}} + (6.7821 - 33.45X_{\text{C}})T \quad (3)$$

where ΔG^{ch} represents the chemical driving force of martensitic transformation; X_{C} and X_{Mn} represent the mole fraction of C and Mn in austenite, respectively; and T represents the temperature. It could be seen that carbon was the most effective element in regulating the driving force of martensitic transformation. In the present study, Ni addition effectively reduced the carbon content in austenite in ADIs. When Ni addition increased to 4 wt%, the carbon content in austenite decreased from 1.9 wt% to 1.6 wt%, thereby reducing the stability of austenite [23]. Notably, the addition of Ni decreased the nucleation rate of the α phase, which not only coarsened the blocky austenite but also impeded the diffusion of carbon from the α phase into the blocky γ phase. Additionally, Ni-Fe substitution further reduced the carbon diffusion rate, resulting in a pronounced carbon concentration gradient within the blocky austenite. Previous studies have similarly documented a non-uniform carbon distribution in blocky austenite [37,38]. Ultimately, higher Ni content facilitated the formation of larger blocky austenite while exacerbating carbon inhomogeneity. The carbon content in the inner region of blocky austenite was low (~1.2 wt%), which was preferentially triggered martensitic transformation (Fig. 12). Twinning deformation would like to occur in the areas with higher carbon content (Fig. 13). The carbon content in the outer region of blocky austenite was high and thus inhibited the martensitic transformation. Therefore, these austenites with enough stability remained, which was confirmed by fracture structures (Fig. 11). Additionally, the volume fraction and size of the blocky austenite in ADI were increased with increasing Ni addition, which was beneficial for reducing its stability [35]. Consequently,

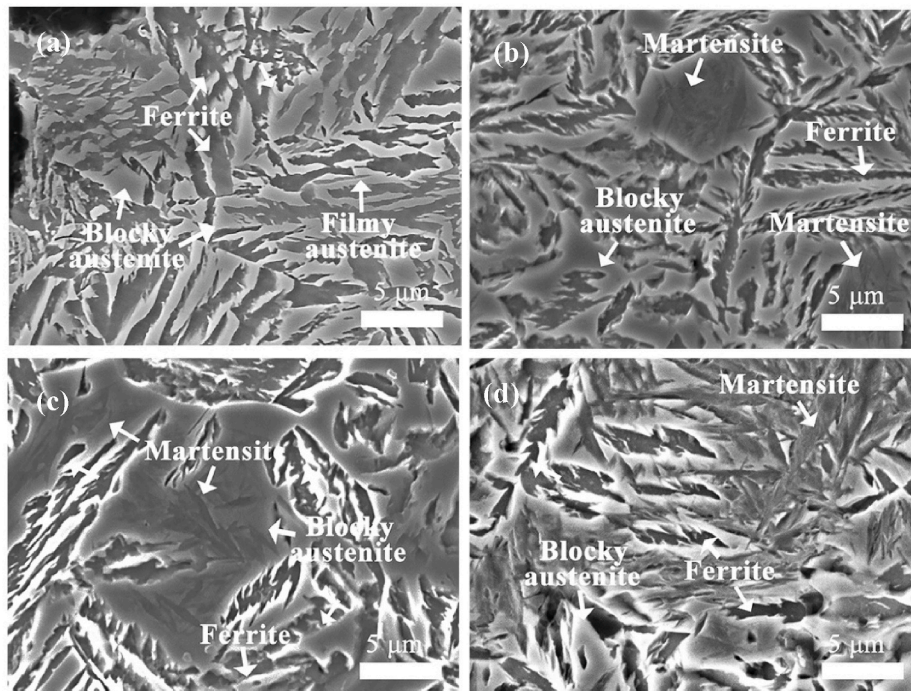


Fig. 11. SEM images of the (a) 0Ni, (b) 2Ni, (c) 4Ni, and (d) 6Ni samples after tensile tests.

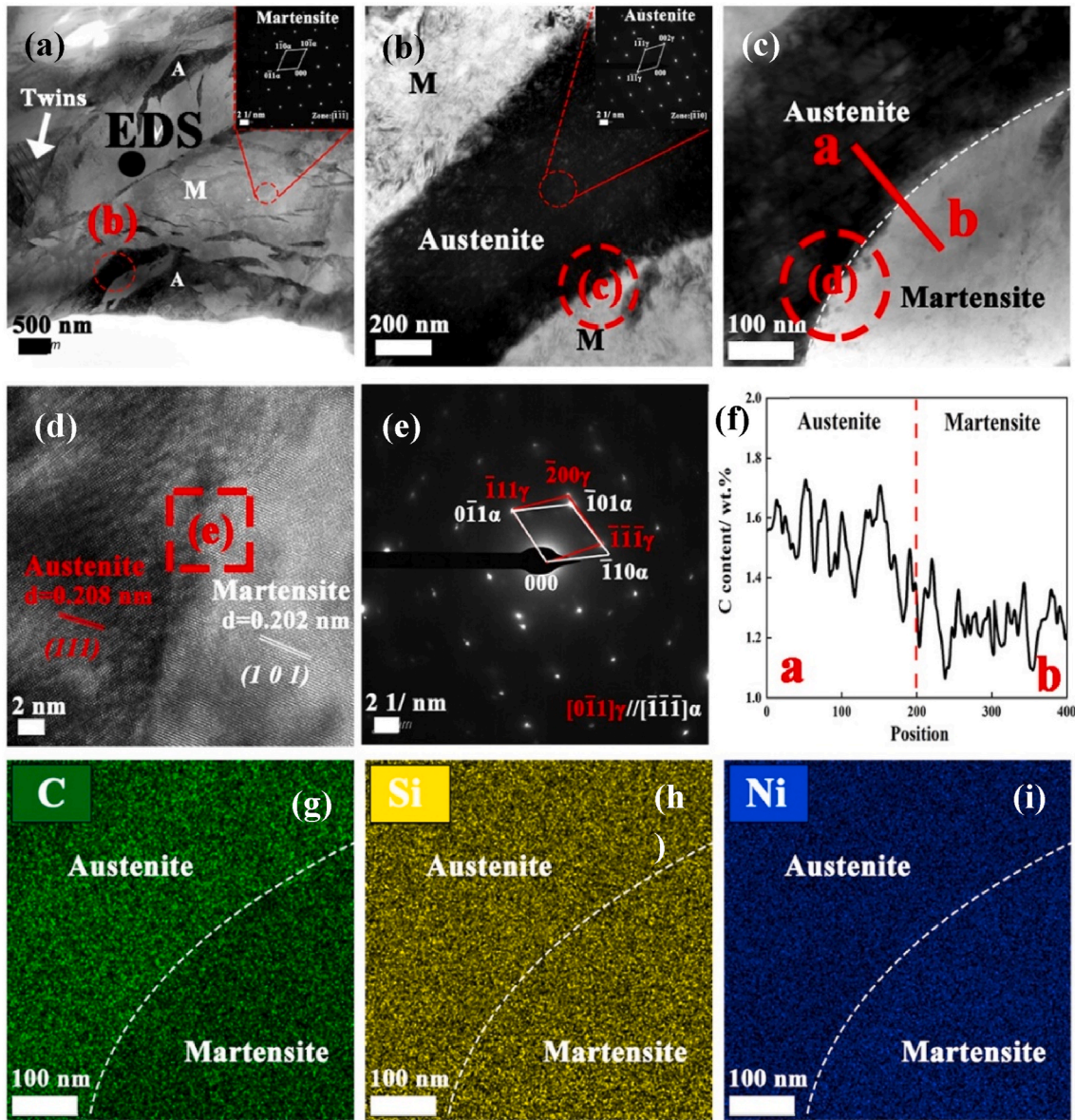


Fig. 12. (a–c) Bright TEM, (d) HRTEM images, and (e) SAED patterns of 4Ni sample after fracture; (f) the distribution of C in martensite and austenite in (c); (g–i) the EDS mapping in (c) showing the distribution of C, Si, and Ni.

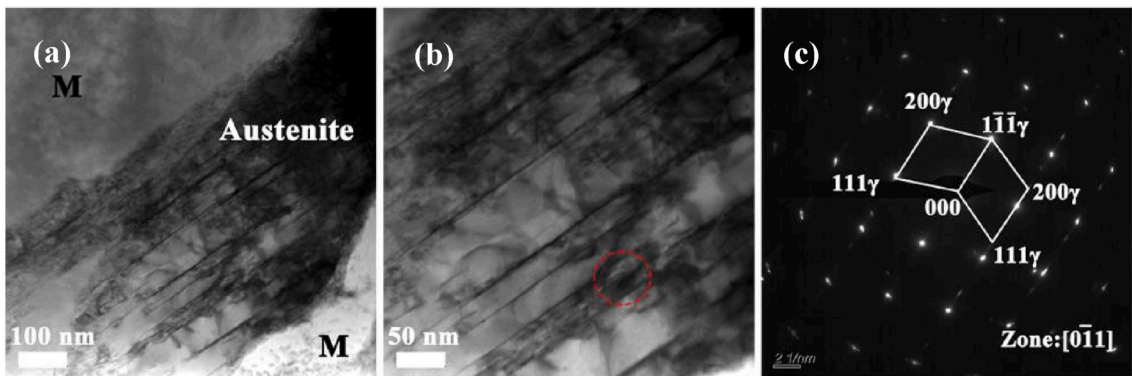


Fig. 13. (a–b) TEM images of deformation twins in the 4Ni sample after fracture and its corresponding (c) SAED.

the amount of strain-induced martensite increased with increasing Ni content.

Additionally, although the carbon content in the austenite of the 6Ni sample was relatively low (Fig. 3), the austenite did not transform into martensite during deformation. This was because the austenite in the 6Ni sample was surrounded by martensite. A large number of defects in martensite hindered dislocation movement and accelerated dislocation accumulation, leading to stress concentration and microcracks [39]. Consequently, dislocation activity in its austenite was hindered from reaching the critical strain required for the austenite-to-martensite transformation. In contrast, the 4Ni sample consisted of ausferrite and blocky austenite, in which the ausferrite exhibited relatively low strength compared with martensite. The low structures' strength differences allowed for a more uniform distribution of strain in 4Ni samples. When there was a high dislocation density in its austenite, dislocations served as nucleation sites for martensite, and the strain energy surrounding dislocations provided the driving force for martensite transformation.

The stability of austenite can be estimated using the following equation [40]:

$$f_{\gamma} = f_{\gamma 0} \exp(-k\varepsilon) \quad (4)$$

where $f_{\gamma 0}$ represents the initial γ phase fraction; f_{γ} represents the γ phase fraction at strain ε , and k represents the stability of the γ phase. A high value of k corresponds to a high driving force for transformation and low stability. In this study, when the Ni content increased from 0 wt% to 2 wt% and 4 wt%, the k was increased from 2.6 to 3.5 and 4.8, respectively, suggesting that the Ni alloying significantly reduced austenite stability. In addition, the k of the 6Ni sample was estimated to be about 2.3, which was lower than the 0Ni sample, indicating that the 6Ni sample showed a higher austenite stability.

Accordingly, appropriate Ni addition significantly reduced the austenite stability and thus enhanced the mechanical properties of ADI through the martensitic transformation. The 4Ni sample demonstrated a good strength-ductility synergy with a UTS of ~ 1100 MPa and a TL of ~ 10 %. Ni alloying increased the austenite fraction and improved ductility. Additionally, the austenite could store a large number of dislocations, and when the dislocation density reached a high level, their strain energy promoted martensitic transformation ($\gamma \rightarrow \alpha'$). This consumed strain energy and alleviated stress concentration, thereby enhancing ductility [32]. Moreover, the newly formed martensite had high dislocation density and elastic modulus, creating hardened regions and hindering dislocation movement. This required a higher applied stress and causing a pronounced rise in the work hardening rate. Subsequent strain accumulation facilitated localized stress relaxation through two key mechanisms: (1) further martensitic transformation of metastable austenite and (2) deformation twinning. These processes accommodated local strain and mitigated stress concentrations, thus reducing the work hardening rate [41]. Consequently, the work hardening rate showed obvious fluctuations (Fig. 8). The amounts of strain-induced martensite increased with increasing Ni addition, giving rise to an increase of working hardening rate [42]. Overall, the above deformation mechanisms enhanced the ADI's work hardening capacity, synergizing with grain refinement to increase ADI's strength. Consequently, the TRIP and TWIP effects simultaneously enhanced the strength-ductility synergy of the 4Ni sample, characterized by a high PSE of ~ 11 GPa%.

5. Conclusions

In the present study, the effects of Ni addition on microstructure evolution and mechanical properties of ADI are investigated. Ni alloying gives rise to the occurrence of strain-induced martensitic transformation, which improves the strength and ductility simultaneously. The main conclusions are summarized as follows.

- (1) Ni alloying alters the microstructural evolution of ductile iron during austempering treatment. With increasing Ni content, the volume fraction and size of blocky austenite increase, while its carbon content decreases. When the Ni content exceeds 6 wt%, the prior austenite transforms into martensite during austempering.
- (2) Strain-induced martensite phenomenon is observed in ADIs containing Ni. The amount of strain-induced martensite increases with increasing Ni content.
- (3) Strain-induced martensitic transformation in ADIs during tensile tests effectively improves work hardening capacity, which results in the simultaneous enhancement of strength and ductility.
- (4) The ADI with 4 wt% Ni achieves an exceptional strength-ductility synergy with a UTS of ~ 1100 MPa and a TL of ~ 10 %, characterized by a PSE of 11 GPa%.

CRedit authorship contribution statement

Xin Wang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuzhou Du:** Writing – review & editing, Supervision, Resources, Conceptualization. **Ziyu Wang:** Writing – review & editing, Resources. **Haohao Li:** Investigation, Formal analysis. **Yanyin Zheng:** Validation, Methodology. **Pengchun Li:** Supervision, Methodology. **Tianjian Mi:** Resources, Methodology. **Bailing Jiang:** Supervision, Resources, Project administration.

Originality statement

I write on behalf of myself and all co-authors to confirm that the results reported in the manuscript are original and that neither the entire work nor any of its parts have been previously published. The authors confirm that the article has not been submitted to peer review, nor has been accepted for publishing in another journal. The author(s) confirm that the research in their work is original and that all the data given in the article are authentic. If necessary, the article can be recalled, and errors corrected.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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